

UP Board 2020

Grade 12

Biology

Time: 3 Hours 15 Minutes

Max. Marks: 70

Note: First 15 minutes are allotted for the candidates to read the question paper.

Instructions :

- i) All questions are compulsory.
- ii) Illustrate your answers with labelled diagrams wherever necessary.
- iii) Marks allotted to each question are mentioned against it.

1 Choose the correct option and write in your answer-book :

A) Algae reproduce asexually by

- i) Conidia
- ii) Buds
- iii) Zoospores
- iv) none of these.

1

B) The development of wale takes place It honeybee from

- i) fertilized egg,
- ii) unfertilized egg
- iii) both (i) and (ii)

iv) none of these.

1

C) Chipko movement was started in

i) Assam

ii) Maharashtra

iii) Garhwal

iv) Bangalore.

1

D) Genetic Engineering is used in

i) Agriculture

ii) Medicine

iii) Vaccine production

iv) All of these

1

2. A) Define pollination.

1

B) What is the name of DNA developed by fusing foreign DNA and vector DNA ?

1

C) Write down the function of stem cells.

1

D) What is formed by interconnection of food chains ?

1

E) Who propounded the chromosomal theory of inheritance ?

1

3. A) Draw a labelled diagram of double helical model of DNA presented by Watson and Crick.

2

B) Which gases were used by Stanley Miller in his

- experiment ? 2
- C) What is standing crop ? 2
- D) Write down the names of trophic levels in the following food chain :
Grass → Goat → Man 2
- E) Write down two main benefits of Apiculture to human beings. 2
- 4 A) Describe Mendel's law of segregation giving example. 3
- B) Describe in brief budding in yeast and sporogenesis in Amoeba. $1\frac{1}{2} + 1\frac{1}{2}$
- C) What is pathogen ? Write down the names of pathogens causing typhoid fever and common cold in man and also symptoms of these diseases. $1 + 1 + \frac{1}{2} + \frac{1}{2}$
- D) Differentiate between the following : $1\frac{1}{2} + 1\frac{1}{2}$
- i) Primary and secondary treatment of sewage.
- ii) Chemical fertilizers and biofertilizers.
5. A) Write down following :
short notes on the $1\frac{1}{2} + 1\frac{1}{2}$
- i) Transgenic animal
- ii) Human insulin synthesis.
- B) Differentiate, between the following : $1\frac{1}{2} + 1\frac{1}{2}$

i) Producer and decomposer

ii) Primary succession and secondary succession.

C) How is sex determined in human beings ? 3

D) Explain polymerase chain reaction. 3

6. A) Draw a labelled diagram of the life cycle of malarial parasite. 3

B) What is genetic code ? Write its four main peculiarities. 1 + 2

C) How do biofertilizers increase the fertility of soil ? 3

D) Write down the functions of restriction enzyme and ligase enzyme. $1\frac{1}{2} + \frac{1}{2}$

7. Give an account of female reproductive system in human beings with the help of labelled diagram. 5

OR

Give an account of the need of family Tanning and its methods in India. $2\frac{1}{2} + 2\frac{1}{2}$

8. What are homology and analogy ? Explain them with examples. 1 + 2 + 2

OR

What are adaptive radiations ? Explain with examples. 1 + 4

9. Give an account of Phosphorus cycle of an Ecosystem. **5**

OR

Write an essay on Wild Animals in India. **5**

